

Mohamadali Tofigh

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Postdoctoral researcher in Computing Science at the University of Alberta, working on reinforcement learning and multimodal foundation models for embodied agents, including LLM-based reward shaping, spatial reasoning in VLMs for motion planning, and foundation-model-assisted teacher–student frameworks for sample-efficient RL. Doctoral background in deep learning for dynamical physical systems, with 12 journal publications and hands-on experience in real-world ML deployment.

EDUCATION

Ph.D. in Mechanical Engineering

Sept 2021 – Mar 2026

University of Alberta

Alberta, Canada

- Thesis: Deep learning for modeling and anomaly detection in fuel cell technology.

ACADEMIC AWARDS

CSME Gold Medal Award - PhD Level

June 2026

Canadian Society for Mechanical Engineering

Canada

- CSME Gold Medals are presented to top-performing students in Mechanical Engineering.

International Society of Automation Scholarship (\$3,000)

May 2026, May 2025

ISA

The United States

- Awards university students globally with outstanding potential for automation and control.

Future Energy Systems Graduate Student Award

Oct 2025

University of Alberta

Alberta, Canada

- Provides support to graduate students to enhance their research.

Alberta Graduate Excellence Scholarship (\$12,000)

Sept 2024, Sept 2025

University of Alberta

Alberta, Canada

- Recognizes the outstanding academic achievement of students pursuing graduate studies in Alberta.

Mojgan Daneshmand Pedram Mousavi Flight PS752 Scholarship (\$20,000)

Aug 2025

University of Alberta

Alberta, Canada

- Awards one graduate student in the Engineering Faculty at the University of Alberta based on superior academic achievement.

Andrew Stewart Memorial Graduate Prize (\$5,000)

June 2025

University of Alberta

Alberta, Canada

- Acknowledges excellence in research at the doctoral level, outstanding accomplishment, and potential in the pursuit of new knowledge.

GSA Graduate Research Assistant Award

Mar 2025

University of Alberta

Alberta, Canada

- Recognizes graduate students who have demonstrated innovation and versatility at the UofA.

Graduate Student Teaching Award

Mar 2025

University of Alberta

Alberta, Canada

- Values the talent and dedication of graduate students for contributions to teaching and learning.

Alberta Innovates Graduate Student Scholarship (\$31,000)

Sept 2024

University of Alberta

Alberta, Canada

- Enables superior graduate students to succeed in Emerging Technology Areas in Alberta.

Sadler Graduate Scholarship (\$14,000)

Sept 2024

University of Alberta

Alberta, Canada

- awards five full-time students in master's or doctoral programs in Mechanical Engineering.

Student Travel Award

July 2024

IEEE | ASME | IFAC

The United State

- supports students to attend the American Control Conference (ACC), and Modeling, Estimation and Control Conference (MECC).

Iran's National Elite Foundation Award

Sept 2015

University of Tehran

Tehran, Iran

- Supplies financial support to outstanding graduate students across nationwide universities.

RESEARCH EXPERIENCE

Postdoctoral Researcher (AI Agents for embodied systems)

March 2026 – Present

University of Alberta

Alberta, Canada

- Develop LLM-based reward-shaping methods to improve reinforcement learning agents for complex continuous-control tasks.
- Investigate spatial reasoning in vision-language and multimodal language models for robotic motion planning.
- Design teacher-student RL frameworks in which foundation models provide action advice to improve sample efficiency.
- Explore policy retrieval and adaptation using skill libraries for fault-tolerant robotic control.
- Study communication efficiency in LLM-based multi-agent systems.
- Co-supervise four master's students and one PhD student.

Doctoral Researcher (Deep Learning and Intelligent Fault Diagnostics)

Sept 2021 – March 2026

University of Alberta

Alberta, Canada

- Leveraged a two-channel interpretable sequence model in TensorFlow, combining Temporal Convolution and Auto-regressive Exogenous. Achieved a 39% runtime and 35% accuracy improvement over standard neural networks for SOFC performance prediction, enabling real-time control/optimization.
- Introduced a Transformer-based neural forecasting algorithm to diagnose early-stage degradation in fuel cells, enabling proactive maintenance strategies to prevent failures.
- Built a semi-supervised time-series anomaly detection framework using temporal graph neural networks for real-time SOFC condition monitoring, achieving 98.7% accuracy.
- Developed a modular PyTorch library for time-series neural forecasting, including data loading, preprocessing, training, validation, and inference pipelines. Implemented advanced Seq2Seq models from scratch, including Transformer, Informer, Autoformer, FEDformer, N-HiTS, N-BEATSx, and TiDE.
- Collaborated with industry partners, including Cummins, to innovate and enhance project outcomes.

Research Assistant (Robotics and Control)

Sept 2015 – Sept 2020

University of Tehran

Tehran, Iran

- Designed a robust non-integer sliding mode control algorithm to stabilize unmanned two-wheeled bicycles at low forward speeds using an innovative gyroscopic actuator.
- Developed a Lyapunov-based nonlinear tracking control strategy for a novel aerial robot. Mathematically proved the asymptotic convergence of the closed-loop system subject to external disturbance.
- Mentored students to fulfill their research and prepare for Iranian ROBOCUP competitions.

TEACHING EXPERIENCE

Teaching Assistant (Machine Learning Control - MecE 610)

Sept 2022 – Sept 2024

University of Alberta

Alberta, Canada

Fall 2022 | Fall 2023 | Fall 2024

- Delivered five hands-on workshops and two interactive lectures, enabling graduate students to develop advanced deep neural networks for sequence modeling in TensorFlow.
- Designed and implemented course materials, creating Python examples, practical workshops, homework assignments, and exams/quizzes that spanned foundational to advanced machine learning concepts, empowering students to build expertise in AI and Python.
- Promoted practical tools in ML engineering, giving insights into design of experiment for collecting representative data for ML-based modeling of dynamic systems.

Teaching Assistant (Feedback Control Design - MecE 420)

May 2022 – May 2025

University of Alberta

Alberta, Canada

Spring 2022 | Winter 2023 | Spring 2023 | Winter 2024 | Spring 2024 | Winter 2025

- Instructed hands-on control lab sessions, guiding students in implementing linear controllers (e.g., PID) on the Quanser Qube hardware setups to bridge theoretical concepts with real-world applications.

Pedagogical Training (SPARK ENG)

Sept 2023 – Jan 2024

University of Alberta

Alberta, Canada

- Completed a comprehensive pedagogical training program covering active learning techniques, interactive teaching, educator identity formation, EDI, and Bloom's taxonomy. Applied these principles to enhance student engagement and learning outcomes in engineering education.

WORK EXPERIENCE

Machine Learning Resident

June 2025 – Jan 2026

Alberta Machine Intelligence Institute (Amii)

Alberta, Canada

- Developed Machine learning and Deep learning algorithms for real-time failure risk prediction and anomaly detection in Gen 3 feed-mixer trucks, including exploratory data analysis, feature engineering, and model design and evaluation.
- Implemented end-to-end ML workflows using Microsoft Azure SQL and Azure DevOps, exploring efficient deployment pipelines for scalable industrial ML applications.

Work Integrated Learning Opportunities (WILO)

Apr 2025 – June 2025

Alberta Machine Intelligence Institute (Amii)

Alberta, Canada

- Collaborated in Client coaching sessions with AI-powered startups to overcome the challenges of scaling ML technology in their business.

COMMUNITY ENGAGEMENT

Mechanical Engineering Graduate Student Association

Sept 2023 – Sept 2025

University of Alberta

Alberta, Canada

- Vice President Events: organized over 15 academic seminars and social events.
- Vice President External: Graduate student representative and voting member in GSA council.

Conference Organization

Sept 2022 – Sept 2025

University of Alberta

Alberta, Canada

- Participated as a student volunteer in CSME 2022, CEEA 2024.
- Served as Symposium Chair in FEGRS 2025.

Scientific Presentation

Sept 2024 – Sept 2025

2024 American Control Conference (ACC)

Toronto, Canada

- Presented innovative contributions to an audience of 150+ scholars.

2025 International Conference on Control, Automation and Diagnosis (ICCAD'25)

Barcelona, Spain

- Delivered an oral presentation of the most recent research results.

Professional memberships

Sept 2024 – Present

- Student Member: Institute of Electrical and Electronics Engineers (IEEE).
- Student Member: American Society of Mechanical Engineers (ASME).
- Student Member: International Society of Automation (ISA).
- Member-in-Training: Association of Professional Engineers and Geoscientists of Alberta (APEGA).

Peer Reviewer Contributions

Sept 2020 – Present

- Elsevier: Applied Energy [IF 11], Engineering Applications of Artificial Intelligence [IF 8], ISA Transactions [IF 7], Robotics and Autonomous Systems [IF 5].
- IEEE: IEEE Control Systems Letters [IF 4], IEEE Robotics and Automation Letters [IF 4.6].

LEADERSHIP

Faculty of Engineering Graduate Research Symposium

Aug 2025

University of Alberta

Alberta, Canada

- Symposium Chair: Collaborating on technical program development and execution, chairing the session on machine learning for engineering.

Soccer Team Captain

Sept 2023 – Jan 2025

University of Alberta

Alberta, Canada

- Led a team of research peers in the Intramural Soccer League, fostering teamwork, communication, and collaboration in a competitive environment.

Team Project Leader

Sept 2015 – Sept 2020

University of Tehran

Tehran, Iran

- Led research teams across major projects, successfully fabricating robots, designing mechatronic systems, and implementing control algorithms in a laboratory environment.

TECHNICAL SKILLS

Programming: Python, MATLAB, SQL, R

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, PyTorch Lightning, MLflow

Cloud/MLOps: Azure DevOps, Docker, Git, AWS, Google Cloud Platform

Methods: Multimodal Large Language Models, Reinforcement learning, Multi-agent AI systems, time-series forecasting, anomaly detection, neural forecasting

SELECTED PUBLICATIONS

Journal Papers:

- [1] Mohamadali Tofigh, M. Fakouri, D.J. Smith, A.R. Hanifi, C.R. Koch, M. Shahbakhti, *RedoxFormer: A Two-Stage Deep Neural Diagnosis Framework for Early-Stage Degradation in Fuel Cells via In-Situ Sensor Measurements*, Applied Energy, Volume 413, 2026.
- [2] Mohamadali Tofigh, D.J. Smith, A.R. Hanifi, C.R. Koch, M. Shahbakhti, *Dynamic Kolmogorov–Arnold Network for Modeling Time-Varying Degradation Patterns in Solid Oxide Fuel Cells*, Advanced Engineering Informatics, Volume 70, 104233, 2026.
- [3] Mohamadali Tofigh, D.J. Smith, A.R. Hanifi, M. Shahbakhti, *A Parallel Seq2Seq Neural Architecture for Long-Horizon Performance Forecasting and Online Condition Monitoring of Fuel Cells*, Applied Soft Computing, Volume 189, 114489, 2026.
- [4] Mohamadali Tofigh, D.J. Smith, C.R. Koch, M. Shahbakhti, *Temporal Dilated Convolution and Nonlinear Autoregressive Network for Predicting Solid Oxide Fuel Cell Performance*, Engineering Applications of Artificial Intelligence, Volume 136, 2024.
- [5] Mohamadali Tofigh, Z. Salehi, A. Kharazmi, D.J. Smith, A.R. Hanifi, C.R. Koch, M. Shahbakhti, *Transient Modeling of a Solid Oxide Fuel Cell Using an Efficient Deep Learning HY-CNN-NARX Paradigm*, Journal of Power Sources, Volume 606, 2024.
- [6] Mohamadali Tofigh, M. Fakouri, D.J. Smith, A. Kharazmi, A.R. Hanifi, C.R. Koch, M. Shahbakhti, *Control-Oriented Modeling of a Solid Oxide Fuel Cell Affected by Redox Cycling Using a Novel Deep Learning Approach*, ASME Journal of Dynamic Systems, Measurement, and Control, Volume 147, No. 2, 2025.
- [7] A. Yasami, Mohamadali Tofigh, M. Shahbakhti, C.R. Koch, *A Generative Physics-Informed Reinforcement Learning-Based Approach for Construction of Representative Drive Cycle*, Transportation Research Part D, Volume 151, 105150, 2026.
- [8] A. Yasami, Mohamadali Tofigh, B. Bahri, C.R. Koch, M. Shahbakhti, *Intelligent Ensemble Architecture for Capturing Transient Nitrogen Oxides (NO_x) Emission Spikes in Real-World Driving Conditions*, Engineering Applications of Artificial Intelligence, Volume 158, 2025.
- [9] Z. Salehi, Mohamadali Tofigh, A. Kharazmi, D.J. Smith, A.R. Hanifi, C.R. Koch, M. Shahbakhti, *Transfer Learning-Based Deep Neural Network Model for Performance Prediction of Hydrogen-Fueled Solid Oxide Fuel Cells*, International Journal of Hydrogen Energy, Volume 99, pp. 102–111, 2025.
- [10] Mohamadali Tofigh, M.J. Mahjoob, M.R. Hanachi, M. Ayati, *Fractional Sliding Mode Control for an Autonomous Two-Wheeled Vehicle Equipped with an Innovative Gyroscopic Actuator*, Robotics and Autonomous Systems, Volume 140, 2021.
- [11] Mohamadali Tofigh, M.J. Mahjoob, M. Ayati, *Dynamic Modeling and Nonlinear Tracking Control of a Novel Modified Quadrotor*, International Journal of Robust and Nonlinear Control, Volume 28, 2018.

Peer-reviewed Conference Papers:

- [1] M. Jafarpour, Mohamadali Tofigh, O. Zaiane, *SAVER: Spatially-Adaptive Visual Reasoning with Self-Reward in Multimodal LLMs*, 41st Annual AAAI Conference on Artificial Intelligence, 2027 (Submitted).
- [2] A. Khan, Mohamadali Tofigh, S. Golestan, O. Zaiane, *AURORA: Adaptive Unified Reward Optimization and Reasoning Approximation*, 41st Annual AAAI Conference on Artificial Intelligence, 2027 (Submitted).
- [3] Mohamadali Tofigh, D.J. Smith, A.R. Hanifi, M. Shahbakhti, *Graph-Attention-Based Multi-Fault Isolation in Fuel Cell Systems*, IFAC Modeling, Estimation and Control Conference (MECC), 2026 (Accepted).
- [4] Mohamadali Tofigh, D.J. Smith, S. Hanifi, A.R. Hanifi, M. Shahbakhti, *A Deep Conv-Atten-KAN Architecture to Identify Solid Oxide Fuel Cell Dynamics under Fuel Starvation*, International Conference on Control, Automation and Diagnosis (ICCAD), IEEE, 2025.
- [5] Mohamadali Tofigh, Z. Salehi, D.J. Smith, A. Kharazmi, A.R. Hanifi, C.R. Koch, M. Shahbakhti, *Developing a Time-Efficient Model for SOFCs Using Self-Supervised Convolutional Autoencoder and Stateful LSTM Network*, American Control Conference (ACC), IEEE, 2024.
- [6] N. Nikafooz, M.J. Mahjoob, Mohamadali Tofigh, *Design, Modeling, and Fabrication of a 3-DOF Wrist Rehabilitation Robot*, 6th RSI International Conference on Robotics and Mechatronics (ICRoM), IEEE, 2018.